

EDA: Canonical Dataset Restructure and Analysis

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1 Overview

This report presents exploratory data analysis of the restructured canonical dataset for the MycoAI fungal species project. The dataset has been reorganized into two named source collections — `curated_primary` and `incoming_low_quality` — each with defined quality tiers and intended uses. A unified preparation pipeline transforms raw source images into a canonical derived hierarchy under `Dataset/prepared/`, organized by species, strain, and environment.

The report covers: source collection profiles, species and environment distributions, metadata quality assessment, the preparation pipeline architecture, and the canonical hierarchy contract.

2 Source Collections

The dataset root now contains two named source collections:

- `curated_primary`: High-quality reference images with complete metadata (species, strain, environment, angle). Used for retrieval holdout evaluation, YOLO training, and fine-tuning.
- `incoming_low_quality`: Lower-quality images from diverse species with partial or missing metadata. Used for generalization testing and diverse data validation.

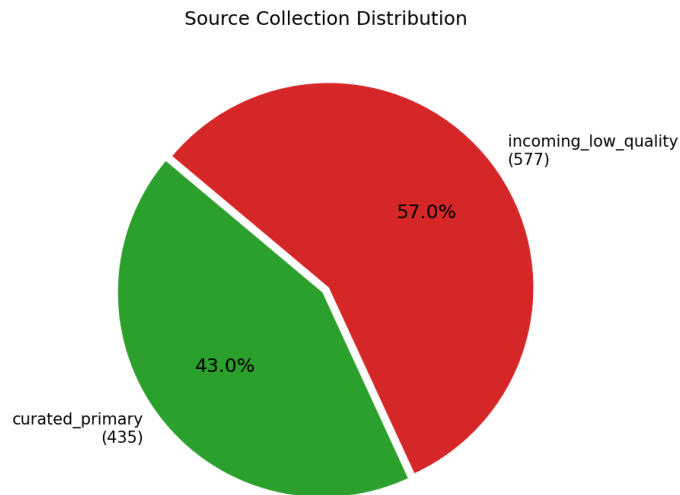


Figure 1: Source collection distribution: `curated_primary` (435) vs `incoming_low_quality` (577)

2.1 Collection Summary

Metric	curated_primary	incoming_low_quality
Images	435	577
Species	8	65
Strains	31	1*
Environments	7	1*
Parse success	100%	0%

Table 1: Collection statistics. (*) Incoming metadata is unresolved — all strains and environments fall back to **unknown**.

3 Species Distribution

3.1 Curated Collection (8 species)

The curated collection contains 8 *Penicillium* species with 4–7 strains per species (except *P. cyclopium* with 1 strain). This collection is the primary source for strain holdout experiments in retrieval evaluation.

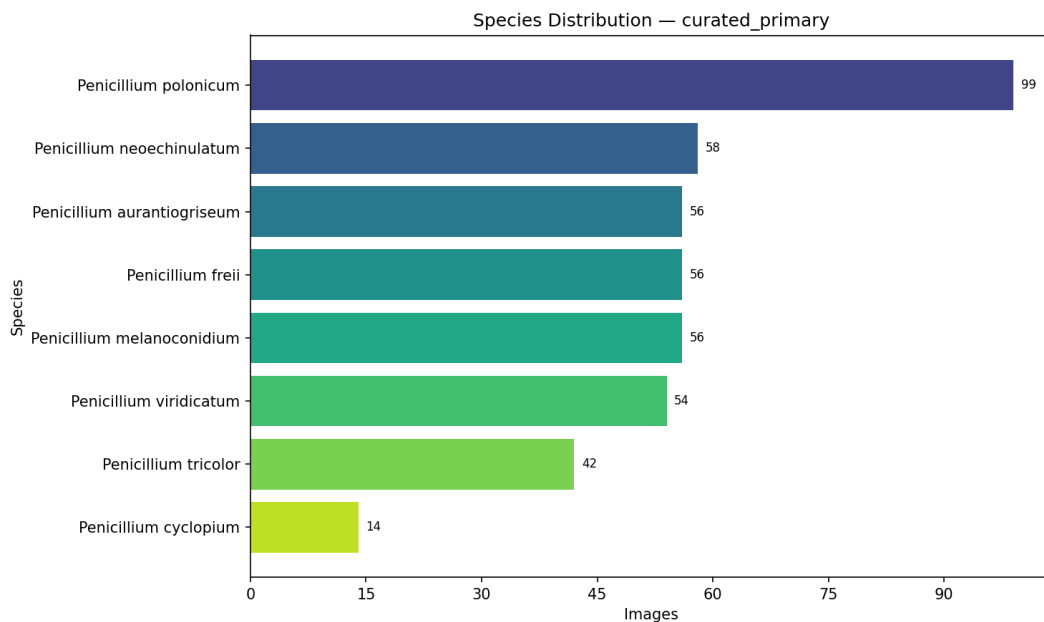


Figure 2: Species distribution — `curated_primary`

3.2 Incoming Collection (65 species)

The incoming collection contains 65 fungal species, mostly from the *Aspergillus* and *Penicillium* genera, with 1–2 strains per species and 4–12 images per species.

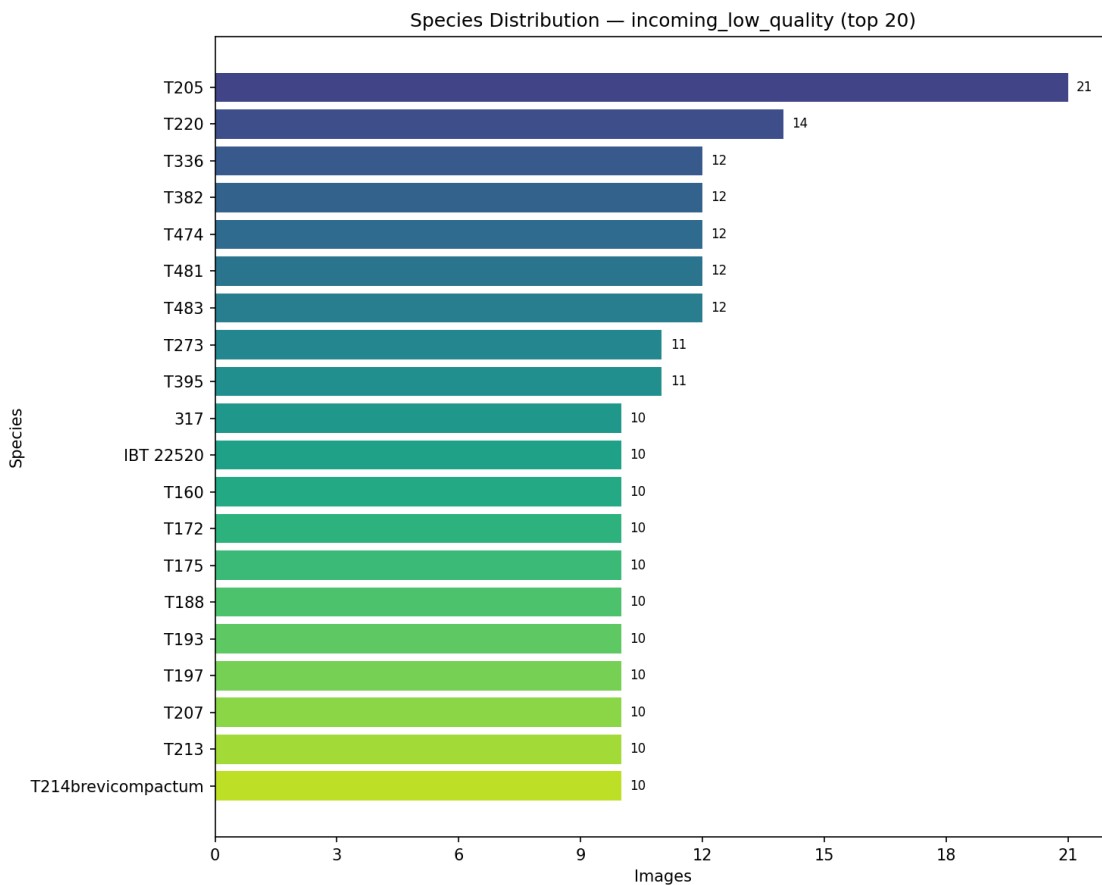


Figure 3: Species distribution — `incoming_low_quality` (top 20)

4 Environment Distribution

The curated collection spans 7 culture environments (CYA, MEA, YES, DG18, CREA, CYA30, CYAS), each with 60–64 images — well-suited for environment-invariant feature learning. The incoming collection currently registers all 577 images as `unknown` environment due to non-standard filename patterns.

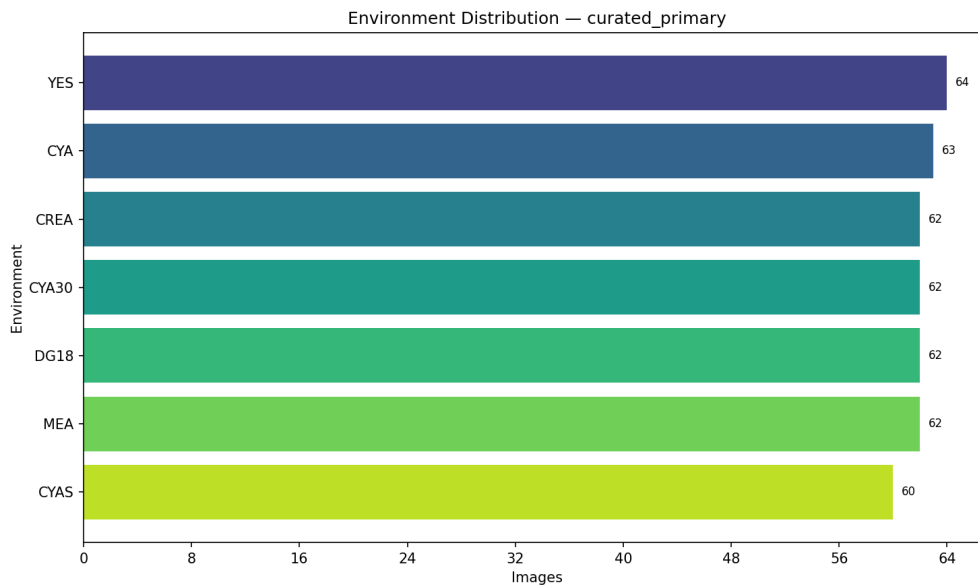


Figure 4: Environment distribution — `curated_primary`

5 Strain Distribution

The curated collection contains 31 unique strains across 8 species. Most strains have 14–16 images each, providing sufficient data for leave-one-strain-out evaluation protocols.

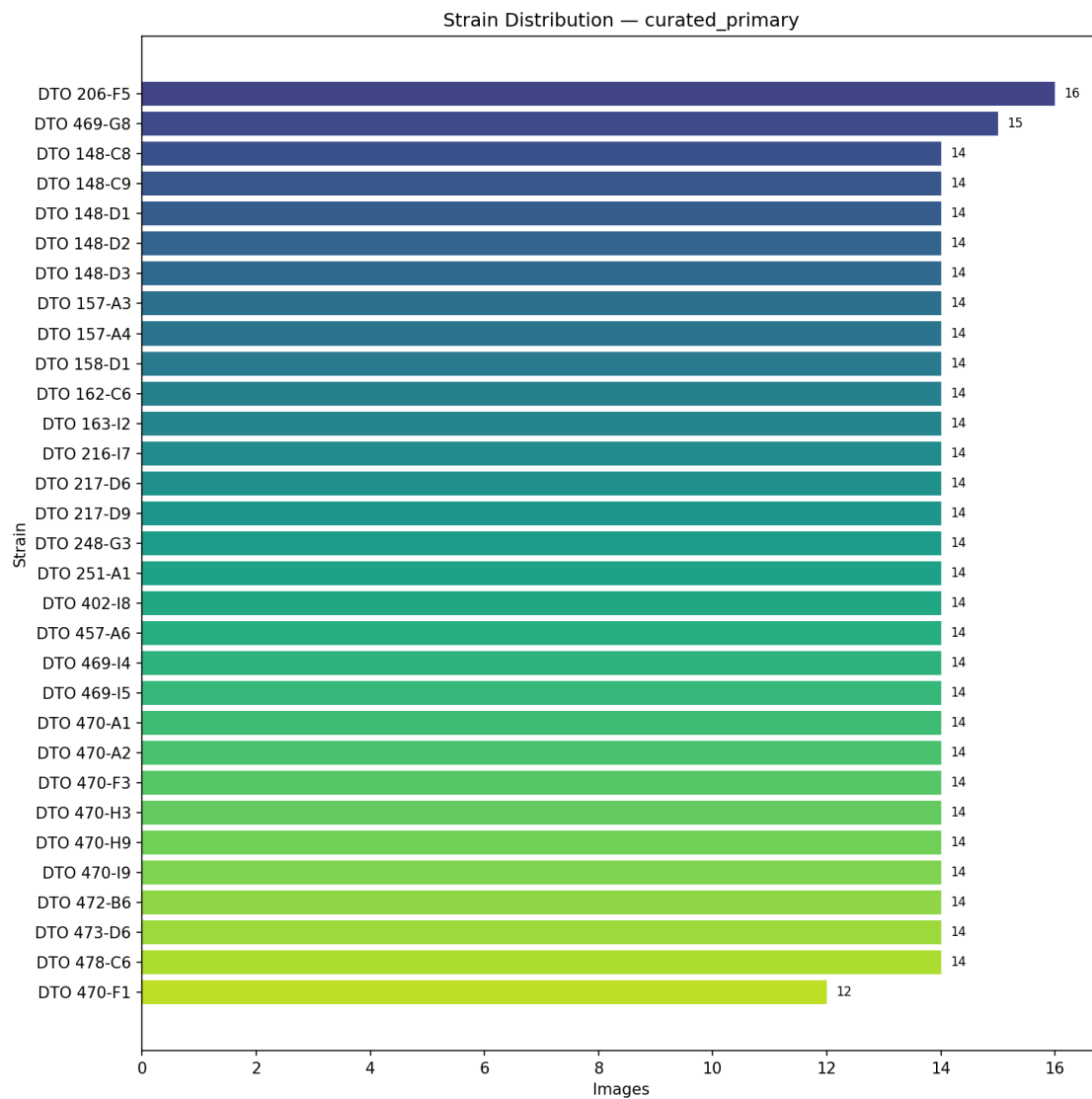


Figure 5: Strain distribution — curated_primary

6 Metadata Quality

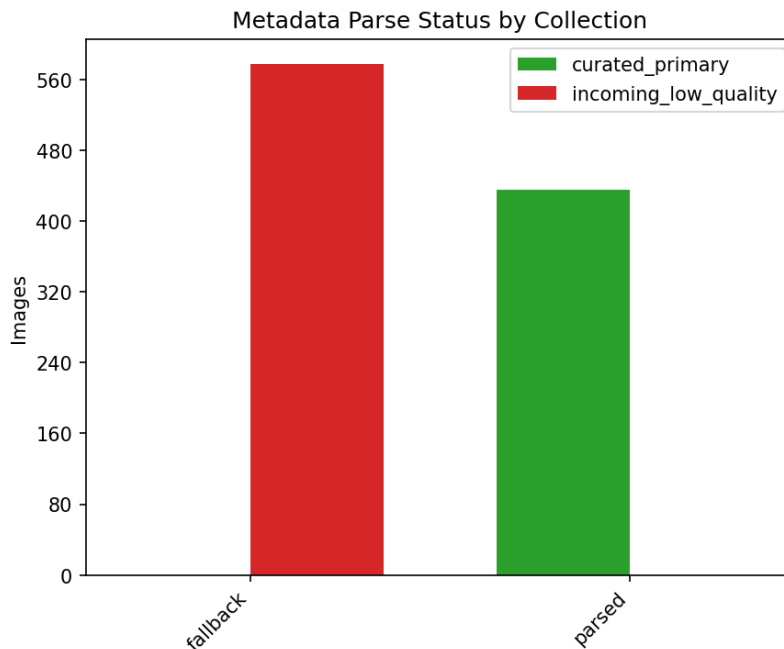


Figure 6: Metadata parse status by collection

All 435 curated images parse successfully via structured filename patterns (e.g., `DT0 148-D1 MEAob`). All 577 incoming images fall through to folder-name-based species inference due to non-standard naming. This is expected behavior given the `incoming_low_quality` designation.

7 Preparation Pipeline

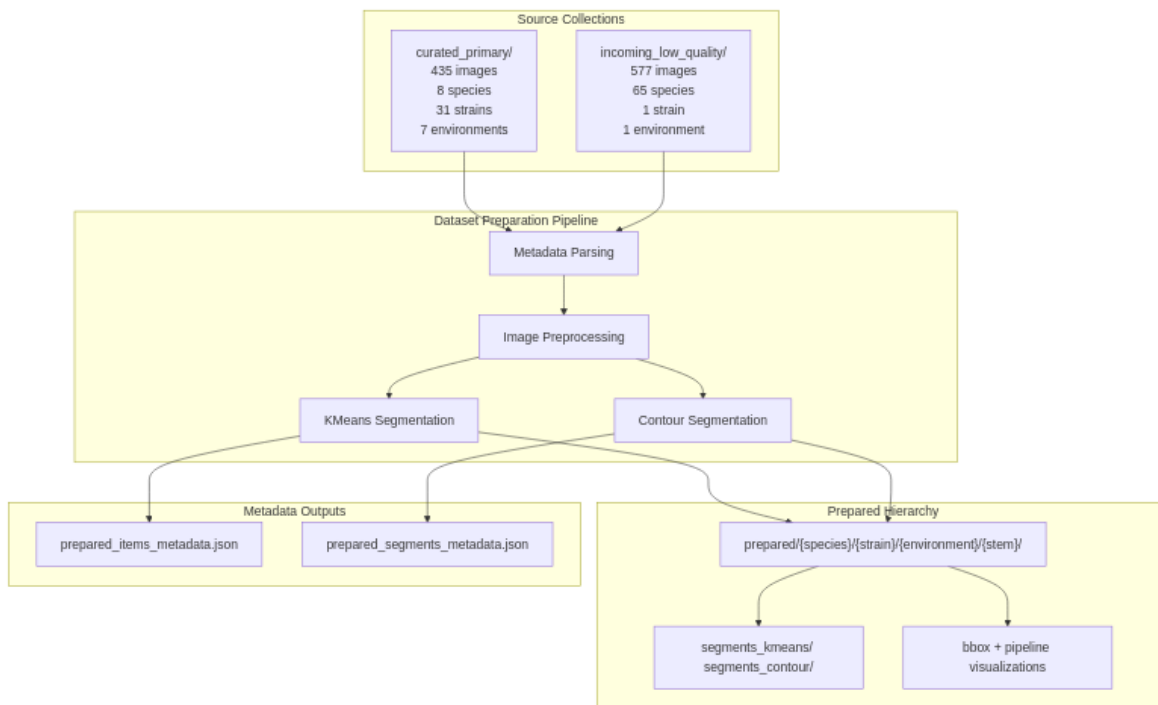


Figure 7: Dataset preparation pipeline: source collections → metadata → preprocessing → segmentation → prepared hierarchy

The unified preparation pipeline in `src/prepare/dataset.py` replaces the legacy `reformat_dataset.py` and `reformat_dataset_yolo.py` scripts. Key features:

- **Single entrypoint:** One script handles both source collections
- **Dual segmentation:** KMeans and contour-based methods for every image
- **Canonical output:** Organized as `prepared/{species}/{strain}/{environment}/{stem}/`
- **Path-authoritative metadata:** Consumers read `segment_path` from JSON metadata instead of constructing flat paths
- **Visualization:** Bounding-box overlays and pipeline visualizations for both segmentation methods

8 Canonical Hierarchy Contract

Path Component	Description
source.jpg	Original source image copy
prepared.jpg	Preprocessed (resized, normalized) image
item.json	Per-image metadata record
segments_kmeans/	KMeans segmentation output (3 segments)
segments_contour/	Contour segmentation output (3 segments)
bbox_*.jpg	Bounding-box overlay visualization
pipeline_*.jpg	Full pipeline visualization

Table 2: Canonical prepared item directory contents

9 Key Findings

1. **Total dataset:** 1,012 images across 73 species, 32 strains, and 8 environments
2. **Curated collection** (435 images) is metadata-rich and suitable for controlled evaluation; all 7 environments are well-represented
3. **Incoming collection** (577 images) brings species diversity (65 species) but requires metadata enrichment before it can support environment-aware experiments
4. **Strain imbalance:** *Penicillium cyclopium* has only 1 strain (14 images) vs 4–7 strains for other curated species — this is a known limitation for retrieval holdout protocols
5. **Canonical hierarchy** is operational: prepared/ output follows {species}/{strain}/{environment}/{step} structure
6. **Legacy fallback:** Dataset/original/ and Dataset/new_data/ still work when canonical curated_primary/ and incoming_low_quality/ paths are absent

10 Environment Configuration

Dataset paths are configurable via environment variables:

Variable	Default	Description
MYCOAI_ROOT	Auto-detected	Monorepo/workspace root
DATASET_ROOT	\$MYCOAI_ROOT/Dataset	Shared dataset folder
WEIGHTS_DIR	\$MYCOAI_ROOT/weights	Model checkpoints
RESULTS_DIR	\$MYCOAI_ROOT/results	Experiment outputs

Table 3: Configurable path environment variables

Set DATASET_ROOT to an external path (e.g., SSD mount, Vast.ai volume) while preserving the canonical folder structure.

11 Conclusion

The dataset restructure from flat legacy paths to a canonical `curated_primary / incoming_low_quality / prepared` hierarchy establishes clear provenance, quality tiering, and predictable downstream consumption patterns. The unified preparation pipeline eliminates code duplication between the old `reformat_dataset.py` and `reformat_dataset_yolo.py` scripts while adding contour segmentation as a second method alongside KMeans. All downstream code paths (feature extraction, Qdrant upload, retrieval evaluation) have been updated to consume metadata-provided `segment_path` fields rather than constructing flat paths from `Dataset/segmented_image/`.